Resolution Geometry: Observer Bounds

What a Limited Observer Can Register, Reach and Predict, with Independent Data Witnesses and Open Predictions

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Abstract

Resolution Geometry (RG) starts from an observer with finite resources: finitely many distinguishable records, finite reach and finite time. This print asks whether that premise does physical work or is only a philosophical frame. It answers in three parts, each backed by a seeded battery in which every group carries a canary. First, a dynamics map records which laws RG derives relative to its premises, which it identifies with established theorems, and which it does not reach. Second, five observer ceilings are placed in one structure: Cramér–Rao (quantum and stochastic), Landauer, Bekenstein, the causal cone with the cosmological event horizon, and the resolution horizon of chaotic cascades. Each ceiling bounds a count of distinguishable outcomes by a budget. At a de Sitter horizon two of the budgets meet in the exact identity $E = TS$: the vacuum energy inside the horizon equals the Landauer cost of erasing the horizon capacity at the horizon temperature. This is a known holographic equipartition, here identified rather than discovered. Third, the premises are confronted with independent data. The measured cosmological constant grounds finite record capacity (about $10^{122.7}$ bits), conditional on the covariant entropy conjecture. Planck 2018 parameters fix the event horizon at 16.7 Gly comoving ($z = 1.87$), so only 4.7% of the presently observable comoving volume is reachable. Three space dimensions are witnessed without circularity by absolute radiometry of the T^4 law. Four spacetime dimensions are witnessed by gravitational data alone: two tensor polarizations, and $D = 4.02^{+0.07}_{-0.10}$ from GW170817 damping. The action quantum agrees across atomic recoil and QED to 10^{-10} and across gravitationally bound neutrons to about 1.5%. An audit withdraws one earlier claim: the COBE/FIRAS spectrum is not a dimension witness, because its calibration assumes Planck's law. Data also select among RG's own constructions: a sequentially composed three-dimensional walk at the Planck spacing has a linear, direction-dependent Lorentz violation. The Fermi GRB 090510 bound excludes it for 26% of lattice orientations from that single burst, while a palindromic walk is untouched. The open predictions are stated with their falsifiers: no propagation-origin linear Lorentz violation (with an octupolar sky pattern if it existed), convergence of the atom-recoil fine-structure determinations, an accuracy ceiling $\Delta S_{\text{tick}}/2k_B$ for discrete-state clocks, and a decade-gain ratio $10^{(p-3)/2}$ for predictability horizons.

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1 Scope and status vocabulary

The companion prints derive results relative to explicit premises. They cover the Einstein branch and its de Sitter attractor (*Physical Dynamics*), the source tensor, signature and dimension (*Tensor Derivation*, second edition), coherence and the task lock of quantum metrics (*Coherence Geometry*), and the conditional low-energy quantum gravity with a Dirac walk and thermal radiation (*Quantum Gravity*, second edition). This print adds no new axiom. It asks three questions:

1. Which dynamics does RG now reach, and with what status?
2. Do the observer limits RG assumes act on measured dynamics?
3. Which conclusions can existing data settle, and which remain for experiment?

Every statement carries one of the following labels.

derived Follows from RG premises in a companion print; the premises are named.

identified The RG statement coincides with an established theorem, which is cited. Nothing new is claimed.

witnessed Independent measurements, not used to build the statement, agree with it. A circularity check is stated.

prediction A statement that existing or near-term data can still refute. The falsifier is named.

conditional Holds under a stated conjecture or assumption.

open Not reached.

Numbers in this print are produced by the battery `rg_observer_bounds.py` (seed 20261111) from published inputs. Each published input is cited where it is used.

2 The dynamics map

Table 1 answers the first question. Its most important column is the last one: what RG does *not* reach.

Table 1: Dynamics reached by RG. Print abbreviations: PD Physical Dynamics, TD Tensor Derivation, CG Coherence Geometry, QG Quantum Gravity, OB this print.

Dynamics	Status	Content	Witness / limit
Linear gravity, source tensor	DERIVED (TD)	Relabeling response is the Hilbert tensor; one symmetric compensator; linear $G^{(1)}$ selected	cone speed, polarizations (OH)
Einstein branch, Λ attractor	DERIVED (PD)	expanding homogeneous solutions approach de Sitter; scalaron decays	measured $\Lambda > 0$ (OB, OC)
Signature $(1,3)$, $D = 4$	DERIVED (TD) relative to premises III, IV	predictive closure gives $\min(p,q) = 1$; propagation gives $D \geq 4$; stable carriers give $D \leq 4$	radiometry (OF), GW (OH)
Quantum state space	CONDITIONAL (QG)	reconstruction theorem imported; polytope rigidity and composition identity proved	—
Free relativistic matter	DERIVED (QG, 1-D)	Dirac walk: exact cone, free Dirac limit	3-D composition constrained by data (OJ)
Action quantum	DERIVED universality (QG); value is an anchor	exchange balance forces one $\hbar$	recoil, QED, neutrons (OI)
Thermal radiation	DERIVED (QG)	$u \propto T^{n+1}$, Wien roots	absolute radiometry (OF)
Quantum metrology	IDENTIFIED (CG)	task selects SLD for estimation	Braunstein–Caves
Stochastic thermodynamics	IDENTIFIED (OB)	TUR is a Cramér–Rao bound of the Fisher geometry	clocks (OK)
Chaotic prediction	IDENTIFIED (OB)	resolution horizon finite iff $p < 3$	weather limit (OL)
Low-energy quantum gravity	CONDITIONAL (QG)	EFT organization, one-loop magnitudes	BMV entanglement (open)
Not reached	OPEN	particle masses, couplings, gauge group, generations, UV completion, interacting 3+1 walk limit, quantum constraint closure	—

The honest summary is short. RG does not yet predict a new force, a new particle or a new constant. It derives the *form* of known low-energy dynamics from operational premises. It identifies the observer ceilings with known theorems. And, as shown below, it lets data choose

among its own constructions.

3 One structure for the observer ceilings

Definition 3.1 (Resolution budget)**.** A resolution task asks an observer to distinguish N outcomes. A *budget* is a resource that is conserved or monotone along the task: energy E, dissipated entropy Σ, spatial extent R, elapsed time t, comoving reach, or resolved scale. A *ceiling* is an inequality of the form $N \le F(\text{budget})$ that holds for every protocol.

Proposition 3.2 (Five ceilings, one form)**.** *The following are ceilings in this sense (*IDENTIFIED*):*

1. Cramér–Rao. *The variance of an unbiased estimate from ν repetitions obeys* $\mathrm{Var} \ge 1/(\nu F)$*, with F the Fisher information of the task metric. The quantum case uses the SLD metric selected by the estimation task; the Margolus–Levitin rate $2E/\pi\hbar$ bounds orthogonal transitions per unit time [6].*

2. Landauer. *Erasing one bit at temperature T dissipates at least $k_B T \ln 2$ [7]. This has been measured [8].*

3. Bekenstein. *A system of energy E inside radius R stores at most $2\pi RE/(\hbar c \ln 2)$ bits [9].*

4. Causal reach. *Signals and travellers stay inside the cone. With $\Lambda > 0$ the reach is also capped by the event horizon.*

5. Resolution horizon. *For a cascade with spectrum $E(k) \propto k^{-p}$, an error below the resolved scale reaches the largest scale in a time that stays finite as resolution is refined, iff $p < 3$ (Section 6.2).*

The unifying claim is structural and modest. Each item counts distinguishable outcomes and bounds the count by a budget. In RG each count is a property of the predictive quotient: a quotient of finite rank, refined at a finite rate, over a finite reach. The proposition does not derive the five theorems from one axiom. It places them in one ledger, so that they can be compared and combined.

Theorem 3.3 (Horizon energy equals Landauer cost of horizon capacity)**.** *For a de Sitter patch of radius R, let E be the vacuum energy inside the horizon, $T = \hbar c/(2\pi k_B R)$ the Gibbons–Hawking temperature [2] and $S = k_B \pi R^2/\ell_P^2$ the horizon entropy. Then exactly*

$$\frac{E}{k_B T} = \frac{\pi R^2}{\ell_P^2} = \frac{S}{k_B}, \qquad E = TS, \qquad \frac{E}{k_B T \ln 2} = \frac{S}{k_B \ln 2}. \tag{1}$$

The Komar energy $|\rho + 3p|V = 2E$ gives the Smarr form $2TS$. For a Schwarzschild horizon, by contrast, $Mc^2/(k_B T_H) = 2S/k_B$.

Proof. $\rho c^2 = 3c^4/(8\pi G R^2)$ and $V = 4\pi R^3/3$ give $E = c^4 R/(2G)$. Dividing by $k_B T$ gives $\pi R^2 c^3/(G\hbar) = \pi R^2/\ell_P^2$. For Schwarzschild, $r_s = 2GM/c^2$ and $T_H = \hbar c^3/(8\pi G M k_B)$ give the factor 2. Group OA checks both symbolically. □

Remark 3.4. This is Padmanabhan's holographic equipartition in another normalization [3]; it is IDENTIFIED, not new. The RG reading is the one in the title: the energy budget of the observable patch is exactly the Landauer price of its record capacity. For the measured Λ, $R_\Lambda = 17.5\,\mathrm{Gly}$, $T = 2.2 \times 10^{-30}\,\mathrm{K}$ and $S/k_B = 3.3 \times 10^{122}$, or 4.8×10^{122} bits. The Schwarzschild canary shows that the factor 1 is specific to the de Sitter horizon and is not generic numerology.

4 Existence and travel bounds from measured parameters

4.1 Finite record capacity is measured, not assumed

Proposition 4.1 (Capacity from Λ; CONDITIONAL)**.** *Assume the covariant entropy conjecture in the form of Bousso's N-bound [4]: in a universe with $\Lambda > 0$, the observable entropy is at most $N = 3\pi/(\Lambda \ell_P^2)$. Planck 2018 gives $\Omega_\Lambda = 0.6847 \pm 0.0073$ and $H_0 = 67.36 \pm 0.54\, km\, s^{-1}\, Mpc^{-1}$ [1], hence $\Lambda = 1.09 \times 10^{-52}\, m^{-2}$, positive at 94σ within ΛCDM. Then every observer has*

$$\log_{10}(\textit{record capacity in bits}) \leq 122.679 \pm 0.008. \tag{2}$$

This turns RG's finite-capacity premise P1 from an assumption into a consequence of a measurement plus one conjecture. The canary is the limit $\Lambda \to 0^+$: the bound diverges, so in a flat or anti-de Sitter world P1 would not be grounded this way. The uncertainty comes from propagating the Planck errors (group OB).

4.2 Cosmic horizons

Proposition 4.2 (WITNESSED inputs, computed)**.** *For flat ΛCDM with Planck 2018 parameters and radiation from $T_{\mathrm{CMB}} = 2.7255\, K$ and $N_{\mathrm{eff}} = 3.046$, the battery gives:*

- *age* 13.795 *Gyr, against the published* 13.797 ± 0.023 *(transcription check);*
- *particle horizon* 46.14 *Gly comoving;*
- *event horizon* 16.69 *Gly comoving, at redshift* $z = 1.871$*;*
- *reachable fraction of the presently observable comoving volume* $(16.69/46.14)^3 = 4.73\%$.

Matter now beyond $z = 1.87$ will never receive a signal sent today. No traveller with any engine reaches it. These values agree with the standard analysis of cosmic horizons [5]. The canary is the Einstein–de Sitter universe ($\Lambda = 0$): the conformal future grows like $2\sqrt{a}$ without bound and no event horizon exists. The existence bound is therefore an effect of the measured Λ, not of kinematics alone.

4.3 Travel

A photon rocket at constant proper acceleration g that accelerates for half the trip and decelerates for the other half covers $d = 2(c^2/g)(\cosh(g\tau/2c) - 1)$ in proper time τ, at mass ratio $e^{g\tau/c}$. At $g = 9.81\,\mathrm{m\,s^{-2}}$ the trip to Andromeda (2.54 Mly) takes 28.6 years of proper time at a mass ratio of 6.9×10^{12}. The battery checks the closed form against the rocket equation to 6×10^{-12}. Galilean kinematics would give about 3100 years. No mass ratio extends the comoving reach beyond 16.7 Gly.

4.4 Ceilings for a human-scale processor

For 1.4 kg inside 0.1 m dissipating 20 W at 310 K:

- Bekenstein capacity: 3.6×10^{42} bits;
- Margolus–Levitin rate: 7.6×10^{50} operations per second;
- Landauer erasure rate: at most 6.7×10^{21} bits per second, or 1.7×10^{31} bits over 80 years.

These are ceilings, not estimates of what brains do. They bite only on models that exceed them. The battery canary rejects a model claiming 10^{50} stored bits in that volume.

5 Independent witnesses of the structural premises

5.1 Three space dimensions from absolute radiometry

Proposition 5.1 (WITNESSED)**.** *In n space dimensions the radiant exitance of a cavity is $\sigma_n T^{n+1}$ (Quantum Gravity, Theorem on thermal radiation). For $n = 3$, $\sigma = 2\pi^5 k_B^4/(15h^3c^2) = 5.670374 \times 10^{-8}\ W\,m^{-2}\,K^{-4}$ from the exact SI constants. Absolute cryogenic radiometry gave $(5.66967 \pm 0.00076) \times 10^{-8}$ between 233 and 373 K [10], which agrees at 0.93σ. If the exitance scaled as $T^{4+\delta}$, inverting with T^4 would offset radiometric from gas-thermometric temperatures by the factor $(T/T_0)^{\delta/4}$. Agreement to 10 mK at 373 K bounds $|\delta| < 3.4 \times 10^{-4}$, and 3 mK bounds it below 1.0×10^{-4}. Hence $n + 1 = 4$ and $n = 3$; changing n by one is excluded by a factor of about 3000.*

The witness is independent. Since 2019, h is realized by the Kibble balance and k_B by acoustic gas thermometry, neither of which uses thermal radiation.

5.2 Audit: what the cosmic spectrum does not witness

Proposition 5.2 (Circularity canary)**.** *Let an instrument output $r(\nu)[B_n(\nu, T_{\rm sky}) - B_n(\nu, T_{\rm cal})]$ with an unknown response r and a blackbody calibrator at thermometer temperature $T_{\rm cal}$. At $T_{\rm sky} = T_{\rm cal}$ the output vanishes identically, for every n and every r.*

FIRAS is such an instrument. Its null ($T = 2.728 \pm 0.004$ K; rms deviation below 50 ppm of the peak; $|y| < 1.5 \times 10^{-5}$; $|\mu| < 9 \times 10^{-5}$ [11]) certifies that the cosmic background is cavity radiation. It does not certify the law of cavity radiation: any absolute frequency-shape statement rests on a response calibrated with Planck's law. The *Quantum Gravity* second edition used the FIRAS shape as a dimension witness. That inference is withdrawn there and replaced by Proposition 5.1. The battery checks that a differential null vanishes for $n = 2, 3, 4$, while an absolute radiometer separates n through the power ratio at doubled temperature, $\log_2 = n + 1$.

5.3 Four spacetime dimensions from gravity alone

Proposition 5.3 (WITNESSED)**.** *Three gravitational measurements, independent of electromagnetic thermodynamics and of each other, agree with $D = 4$:*

1. Polarization count. *A massless spin-2 field in D dimensions has $D(D-3)/2$ polarizations, which equals 2 only for $D = 4$. The three-detector event GW170814 favours pure tensor polarization over pure vector by a Bayes factor above 200, and over pure scalar by more than 1000 [12].*

2. Amplitude damping. *Comparing the gravitational-wave luminosity distance of GW170817 with the electromagnetic distance of its host gives $D = 4.02^{+0.07}_{-0.10}$ (SH0ES H_0) or $3.98^{+0.07}_{-0.09}$ (Planck H_0) [13]. $D = 5$ is excluded at 14σ, and $D = 3$ at 10σ, by the quoted one-sided widths.*

3. One cone. *GW170817 with GRB 170817A gives $-3 \times 10^{-15} \le v_g/c - 1 \le 7 \times 10^{-16}$ [14].*

These are data *for* premises III and IV of the *Tensor Derivation*. The derivation does not use them, and they do not use it.

5.4 One action quantum across sectors

Proposition 5.4 (WITNESSED; tension recorded)**.** *The fine-structure constant from atom recoil, h/m [15, 16], and from the electron magnetic moment with QED [17] gives:*

Determination	α^{-1}	*pair z*
Rb recoil 2020	137.035999206(11)	*Rb–Cs* +5.49
Cs recoil 2018	137.035999046(27)	*Cs–g*−2 −3.89
electron g − 2 *2023*	137.035999166(15)	*Rb–g*−2 +2.15

The Birge ratio of the triple is 3.9*; none of* 2000 *consistent synthetic triples reaches it. Without Cs it is* 2.2.

Atom-interferometric recoil and QED loops therefore share one $\hbar$ at about 10^{-10} relative, with one unresolved pair. For gravitationally bound neutrons, transition frequencies scale as $g^{2/3}\hbar^{-1/3}$. The qBounce value $g = 9.84 \pm 0.04\,\mathrm{m\,s^{-2}}$ [18] agrees with terrestrial g, which bounds $|\hbar_{\mathrm{grav}}/\hbar - 1|$ at about 1.5%. RG requires one exchange quantum for all carriers (*Quantum Gravity*, exchange balance). The Rb–Cs tension is addressed in Prediction 2.

6 Observer limits that bite measured dynamics

6.1 Clocks: the thermodynamic uncertainty relation is Cramér–Rao

Theorem 6.1 (TUR from the Fisher geometry; IDENTIFIED)**.** *Let a continuous-time Markov jump process on a finite graph have rates w_{ij} and steady state p. Set $f_{ij} = w_{ij}p_j$, the net flux $j_{ij} = f_{ij} - f_{ji}$ and the traffic $a_{ij} = f_{ij} + f_{ji}$. Then for any antisymmetric edge current J with steady-state rate $\langle J\rangle$ and variance rate* $\mathrm{Var}(J)$*,*

$$\frac{\mathrm{Var}(J)}{\langle J\rangle^2} \;\geq\; \frac{1}{I} \;\geq\; \frac{2}{\Sigma}, \qquad I = \sum_{\text{edges}} \frac{j_{ij}^2}{a_{ij}}, \qquad \Sigma = \sum_{\text{edges}} j_{ij} \ln \frac{f_{ij}}{f_{ji}}. \tag{3}$$

Proof. Perturb the rates by $\partial_\theta \ln w_{ij} = j_{ij}/a_{ij}$. The net flux on every edge changes by $j_{ij}(f_{ij} + f_{ji})/a_{ij} = j_{ij}$. Every net flux is thus scaled by $1 + \theta$, the stationarity condition is preserved, and every current mean scales the same way. The Fisher information rate of the path measure at $\theta = 0$ is $\sum_{i\neq j} f_{ij}(\partial_\theta \ln w_{ij})^2 = I$. Cramér–Rao for estimating θ from J gives the first inequality. The log-mean inequality $(x - y)\ln(x/y) \geq 2(x - y)^2/(x + y)$ gives $I \leq \Sigma/2$. □

This is the thermodynamic uncertainty relation [19] in the information-inequality form [20]. RG adds only the reading: precision of a record costs dissipation through the same Fisher metric that RG uses for resolution. Group OK checks $I \leq \Sigma/2$ and both inequalities on 120 random networks; the worst ratio is $1 + 1.7 \times 10^{-6}$. Near equilibrium the bound saturates (ratio 0.9999992 after numerical differentiation), so a claimed constant 4 in place of 2 fails there; that is the canary.

Corollary 6.2 (Clock ceiling)**.** *For a discrete-state clock whose ticks are net crossings of one edge, the accuracy per tick (squared mean over variance of the period) is at most $\Delta S_{\mathrm{tick}}/(2k_B)$.*

The data are consistent with this ceiling. Models of biochemical oscillators lie orders of magnitude below it [21]. A nanomechanical clock driven by noise showed accuracy linear in entropy per tick [22]. That clock is a continuous-variable system, outside the jump-process class, with its own proportionality constant, so it tests the linear law and not the constant 1/2.

6.2 Weather: the resolution horizon

Proposition 6.3 (Resolution horizon of a cascade; IDENTIFIED)**.** *Let shells $k_m = k_0 r^m$ carry kinetic energy $E(k) \propto k^{-p}$, and let an error saturate shell m after the eddy time $\tau_m \propto (k_m^3 E(k_m))^{-1/2} \propto k_m^{(p-3)/2}$. An error confined below the resolved shell M reaches the largest scale after $T(M) = \sum_{m\leq M} \tau_m$. Then:*

1. *$T(M)$ converges as $M \to \infty$ iff $p < 3$;*

2. *per decade of resolution, the gain in horizon shrinks by the factor $10^{(p-3)/2}$; this is 0.215 for $p = 5/3$ and 1 for $p = 3$;*

3. *for a spectrum with slope -3 at large scales and $-5/3$ beyond a transition, the horizon is finite.*

Proof. τ_m is geometric with ratio $r^{(p-3)/2}$. Group OL checks the ratios exactly with $r = 10^{1/3}$. □

This is Lorenz's 1969 argument [23] made explicit. The measured mesoscale atmospheric spectrum has slope $-5/3$ [24]. The consequence is an intrinsic horizon: once the error sits in the $-5/3$ range, no improvement in observation moves the deterministic forecast limit beyond a finite value. High-resolution experiments put the midlatitude limit near two weeks and find that reducing the initial error by an order of magnitude gains up to five days [25]. Here the finite observer is not philosophy: it sets a horizon in a measured system. The canary is $p = 3$, which gives constant gains per decade and no intrinsic limit.

6.3 Lattice Lorentz violation: data choose the walk

Proposition 6.4 (Parity of the leading violation; DERIVED for the stated walks)**.** *1. The one-dimensional Dirac walk of* Quantum Gravity *has $\cos\omega = \sqrt{1-m^2}\cos k$. This is even in k, so its leading violation is quadratic.*

2. *The sequential three-dimensional walk $U = e^{-ik_x\sigma_x}e^{-ik_y\sigma_y}e^{-ik_z\sigma_z}$ has $\cos\omega = c_xc_yc_z - s_xs_ys_z$. Hence $\omega = |k| + f|k|^2 + O(k^3)$ with $f = n_xn_yn_z$, $|f| \le 1/(3\sqrt{3})$. This is a linear, direction-dependent violation with $E_{\mathrm{QG},1} = E_a/(2|f|)$, where $E_a = \hbar c/a$.*

3. *The palindromic walk $e^{-ik_x\sigma_x/2}e^{-ik_y\sigma_y/2}e^{-ik_z\sigma_z}e^{-ik_y\sigma_y/2}e^{-ik_x\sigma_x/2}$ satisfies $U(-k) = U(k)^\dagger$. Hence $\cos\omega$ is even, there is no linear term, and the relative deviation is $O(k^2)$.*

Proof. For (2), $\sigma_x\sigma_y\sigma_z = i$ gives the scalar part of the product. Expanding, $\cos\omega \approx 1 - |k|^2/2 - k_xk_yk_z$, so $\omega \approx |k| + k_xk_yk_z/|k|$. For (3), a palindromic product of exponentials of $-k$ is the inverse of the product at k. The battery fits the linear coefficient on 40 random directions (error below 10^{-4}). It also checks that the palindromic absolute deviation grows by the factor 8 under doubling of k, which is the cubic term. □

Proposition 6.5 (Confrontation with Fermi data)**.** *For GRB 090510 ($z = 0.903$) with the cosmology of the discovery paper, the conservative photon energy $30.53 - 2.56$ GeV and the delay $\Delta t < 0.859$ s give $E_{\mathrm{QG},1} > 1.19E_{\mathrm{Pl}}$. The battery recomputes $1.193E_{\mathrm{Pl}}$ from these inputs, matching the published value [26]. The likelihood analysis later gives $E_{\mathrm{QG},1} > 7.6E_{\mathrm{Pl}}$ (subluminal, 95%) and $E_{\mathrm{QG},2} > 1.3 \times 10^{11}$ GeV [27]. Suppose the sequential walk has $a = \ell_P$ and photons propagate on the same lattice. It is then excluded wherever $f < -1/(2 \cdot 7.6) = -0.066$. For a uniformly random lattice orientation this happens with probability 0.26, from this single burst. The quadratic delay of the same photon at $E_{\mathrm{QG},2} = E_{\mathrm{Pl}}$ is 6×10^{-18} s, which cannot be tested.*

This is the sharpest new statement in the print, and it cuts against one of RG's own options. A sequentially composed three-dimensional walk is disfavoured by existing data, and a population of bursts at different directions would test it further. The palindromic (time-symmetric) composition is not touched. The assumption that photons share the matter lattice is labelled: the quantum-gravity print constructs only a fermionic walk.

7 Predictions and falsifiers

Prediction 1 (No linear propagation delay; octupolar signature if wrong)**.** The physical walk is palindromic, so no energy-dependent arrival delay linear in E and proportional to $K_1(z) = \int_0^z (1+z')\,\mathrm{d}z'/H(z')$ exists. A sequential walk would instead produce delays $\propto n_x n_y n_z\, E\, K_1(z)$. This is an $\ell = 3$ pattern on the sky, with sign changes between octants, tied to the lattice axes. *Test with existing data:* fit published Fermi-LAT burst delays and TeV flare delays to $K_1(z)$ times an $\ell = 3$ cubic harmonic. *Falsifier:* a significant propagation-origin linear term, isotropic or octupolar.

Prediction 2 (Fine-structure convergence)**.** Universal $\hbar$ (one exchange quantum) and universal mass ratios imply that the Rb–Cs recoil difference is systematic. Future recoil determinations will converge, within errors, on each other and on the $g-2$ value. This prediction is shared with standard physics. *Falsifier:* a species-dependent h/m surviving independent systematic checks at more than 5σ.

Prediction 3 (Clock ceiling)**.** No discrete-state (jump-process) clock, biochemical, molecular or engineered, exceeds an accuracy per tick of $\Delta S_{\mathrm{tick}}/2k_B$. *Test with existing data:* published accuracy and dissipation of molecular motors and synthetic oscillators. *Falsifier:* a measured jump-process clock above the ceiling.

Prediction 4 (Diminishing returns in forecasting)**.** In twin or ensemble experiments, successive tenfold reductions of initial error that act in the $-5/3$ range yield horizon gains whose ratio tends to $10^{-2/3} \approx 0.22$. Gains in the -3 range stay approximately constant. *Test with existing data:* error-reduction experiments of the type in [25]. *Falsifier:* constant gains per decade persisting deep into the $-5/3$ range.

Prediction 5 (Dimension at larger distances)**.** Gravitational-wave sirens with electromagnetic counterparts at larger distances will continue to give $D = 4$. Leakage at any screening scale would contradict the dimension theorem of the *Tensor Derivation*, and hence one of its premises III or IV. *Falsifier:* a significant distance-dependent excess damping of gravitational waves relative to light.

Prediction 6 (Medium term: gravitationally induced entanglement)**.** The *Quantum Gravity* print records the entangling phase of the Bose–Marletto–Vedral configuration (negativity 0.078 in its toy parameters). A positive entanglement witness is expected. This is not a near-term test and is listed only for completeness [28].

Table 2: What is new, what is identified, and where it can fail.

Item	Status	Data now	Fails if
$E = TS$ at de Sitter horizon	IDENTIFIED	exact	—
P1 from measured Λ	CONDITIONAL, new in RG	Planck 2018	$\Lambda \leq 0$, or the N-bound fails
Event horizon, reach 4.7%	computed	Planck 2018	parameters shift
$n = 3$ by radiometry	WITNESSED	Quinn–Martin	$T^{4+\delta}$, $\lvert\delta\rvert > 3 \times 10^{-4}$
FIRAS not a witness	correction	—	—
$D = 4$ by gravity	WITNESSED	GW170814, GW170817	excess damping

Item	Status	Data now	Fails if
one $\hbar$	WITNESSED, tension	α triple, qBounce	species-dependent h/m
TUR as Cramér–Rao	IDENTIFIED	oscillators, clocks	jump clock above ceiling
Resolution horizon	IDENTIFIED	weather limit	constant decade gains at $-5/3$
Walk parity selection	new	GRB 090510	linear delay $\propto K_1(z)$

8 Limits of this print

- No new dynamical law is derived here. The new items are a premise grounded by measurement (P1), a selection among RG's own walk constructions by data, a correction of an earlier witness, and testable patterns.
- Proposition 4.1 depends on the covariant entropy conjecture.
- Proposition 6.5 assumes that photons propagate on the matter lattice. It uses a single burst, the subluminal branch only, and a uniform prior on lattice orientation.
- The temperature-agreement level in Proposition 5.1 is taken as 10 mK, which is conservative relative to the quoted few mK.
- The cascade model in Proposition 6.3 is a shell model. Its quantitative transfer to an operational forecast system needs the system's own error-growth calibration.
- The fine-structure prediction is not specific to RG.

A Battery

The battery `rg_observer_bounds.py` needs numpy, scipy and sympy and uses seed 20261111. It writes its results to `rg_observer_bounds_results.json` and exits nonzero on any failure. Six mutations were run against it: shifting the Quinn–Martin value, replacing the palindromic walk by the sequential walk, inflating the Fisher rate, changing the polarization count formula, moving the Cs value onto the Rb value, and forcing $p = 3$ gains. Each mutation is caught by the intended group.

Group	Content and canary
OA	$E/(k_BT) = S/k_B$ exact; Komar $2TS$; Schwarzschild factor 2 canary
OB	capacity from Λ; error propagation; $\Lambda \to 0$ canary
OC	age, particle and event horizons, reach; Einstein–de Sitter canary
OD	photon rocket closed form versus ODE; Andromeda; Galilean canary
OE	Bekenstein, Margolus–Levitin, Landauer budgets; over-capacity canary
OF	σ from constants versus Quinn–Martin; exponent bound; $n = 2, 4$ canaries
OG	differential null independent of n (circularity canary); absolute radiometer separates n
OH	polarization count; Pardo et al. D; cone speed; $D = 5$ canary
OI	α pairs, Birge ratio, synthetic-consistency canary; qBounce $\hbar$ bound
OJ	1-D walk even; 3-D sequential linear coefficient $= n_x n_y n_z$; palindromic even (cubic scaling); GRB 090510 recomputation; orientation probability

OK	TUR via Fisher $\leq \Sigma/2$ on 120 networks; saturation; fake-4 canary; clock ceiling
OL	decade-gain ratios $10^{(p-3)/2}$; two-regime finite limit; $p = 3$ canary

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